

Tutorial

Calculating TF-IDF For Three Documents

Course:

Natural Language Processing

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TF-IDF: Complete Worked Example

We will use **three simple documents**:

D1: the cat sat on the mat

D2: the dog sat on the log

D3: the cat chased the dog

Step 1: Build Vocabulary

List all unique terms:

["the", "cat", "sat", "on", "mat", "dog", "log", "chased"]

Step 2: Term Frequency (TF)

$$TF = (\text{number of times term appears in doc}) / (\text{total terms in doc})$$

Document Lengths

D1: 6 words

D2: 6 words

D3: 5 words

TF Table

Term	TF(D1)	TF(D2)	TF(D3)
the	2/6	2/6	2/5
cat	1/6	0	1/5
sat	1/6	1/6	0
on	1/6	1/6	0
mat	1/6	0	0
dog	0	1/6	1/5
log	0	1/6	0
chased	0	0	1/5

Step 3: Inverse Document Frequency (IDF)

$$IDF = \log(N / df)$$

N = number of documents = 3

df = number of documents containing the term

Natural Log (ln)

Term	df	IDF = $\log(3/df)$
the	3	$\log(3/3)=0$
cat	2	$\log(3/2)=0.176$
sat	2	0.176
on	2	0.176
mat	1	$\log(3)=1.099$
dog	2	0.176
log	1	1.099
chased	1	1.099

Step 4: $TF \times IDF$

Multiply TF and IDF values.

TF-IDF for D1

Term	TF(D1)	IDF	TF-IDF
the	2/6	0	0
cat	1/6	0.176	0.029
sat	1/6	0.176	0.029
on	1/6	0.176	0.029
mat	1/6	1.099	0.183

TF-IDF Vector for D1:

[0, 0.029, 0.029, 0.029, 0.183, 0, 0, 0]

TF-IDF for D2

Term	TF(D2)	IDF	TF-IDF
log	1/6	1.099	0.183
dog	1/6	0.176	0.029
sat/on same as above			

D2 Vector:

[0, 0, 0.029, 0.029, 0, 0.029, 0.183, 0]

TF-IDF for D3

Term	TF(D3)	IDF	TF-IDF
cat	1/5	0.176	0.035
chased	1/5	1.099	0.220
dog	1/5	0.176	0.035

D3 Vector:

[0, 0.035, 0, 0, 0, 0.035, 0, 0.220]

Final Summary**TF-IDF Vectors**

Document	TF-IDF Vector
D1	[0, 0.029, 0.029, 0.029, 0.183, 0, 0, 0]
D2	[0, 0, 0.029, 0.029, 0, 0.029, 0.183, 0]
D3	[0, 0.035, 0, 0, 0, 0.035, 0, 0.220]

Insights from the TF-IDF Example**Most Important Observations**

- Rare words receive higher TF-IDF weights.**
 In our example, terms like "mat", "log", and "chased" appear in **only one document**, so their **IDF is highest (≈ 1.099)**. Even with modest TF, they get strong TF-IDF values.
 → *Example:* "mat" in D1 gets **0.183**, the highest in that document.
- Common terms get very low TF-IDF scores.**
 Words like "the" appear in all documents ($df = 3$), so **IDF = 0**, and therefore **TF-IDF = 0 for all documents**. These words carry almost no semantic value for distinguishing documents.
- TF-IDF highlights content-specific keywords.**
 Words unique to a document (mat, log, chased) emerge as **key identifiers**, showing what that document is actually "about".
- The method balances both frequency and uniqueness.**
 Even if a word occurs many times in one document, TF-IDF won't rank it high unless it is **rare across the corpus**.

How TF and IDF Control Importance

1. Term Frequency (TF)

- TF measures **how frequently a word appears in a single document**.
- It increases importance **within** the document but does **not** consider other documents.
→ Example: "cat" appears in D1 and D3, so TF helps, but IDF limits its importance.

2. Inverse Document Frequency (IDF)

$$\text{IDF} = \log(N / \text{df})$$

Where:

- **N** = total number of documents
- **df** = number of documents containing the term

How IDF Behaves

- **Rare word → lower df → higher IDF → higher importance.**
→ "chased" appears only in D3 → IDF = 1.099 → strong TF-IDF.
- **Common word → higher df → lower IDF → lower importance.**
→ "the" appears in all docs → IDF = 0 → 0 importance everywhere.

Key IDF Behaviors

- **Higher denominator (more docs with the word) → IDF decreases**
- **Lower denominator (fewer docs with the word) → IDF increases**
- The **log** simply stabilizes the scale so values don't explode.

Summary

Concept	Effect
TF (within-document importance)	Increases weight based on frequency in the document
IDF (corpus-wide rarity)	Increases weight when word is rare across all documents
TF × IDF	High if a word is frequent <i>and</i> rare; low if a word is rare <i>or</i> common

TF-IDF therefore gives high importance to **rare yet meaningful words** and suppresses extremely common words that do not contribute to understanding document-specific topics.